

# 8255 Programmable Peripheral Interface (PPI) — MPCA

## Module 4

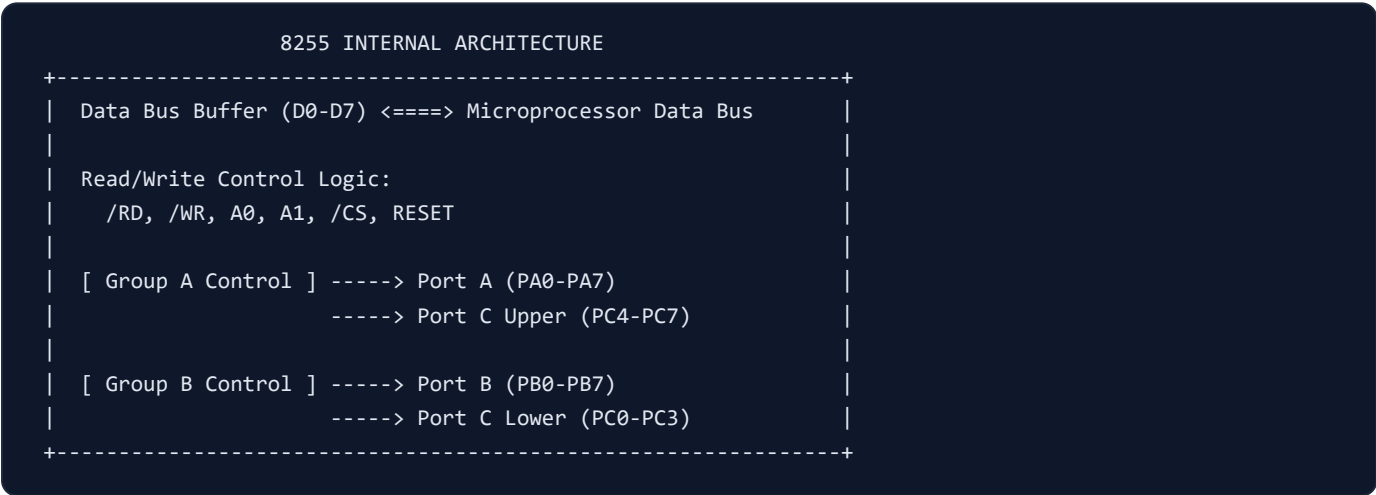
**Definition:** The **Intel 8255A (PPI)** is a general-purpose programmable I/O device designed to interface parallel peripheral devices (keyboards, displays, ADCs, printers) to the 8086 microprocessor.

### 1. Detailed Technical Explanation

#### 1. Internal Block Diagram of 8255

The 8255 features **24 programmable I/O pins** divided into three 8-bit ports:

- **Port A (8 pins - PA0 to PA7):** Can be programmed as 8-bit Input, Output, or Bidirectional port.
- **Port B (8 pins - PB0 to PB7):** Can be programmed as 8-bit Input or Output port.
- **Port C (8 pins - PC0 to PC7):** Can be split into two 4-bit ports: **Port C Upper (PC4-PC7)** and **Port C Lower (PC0-PC3)** for handshake signals or simple I/O.

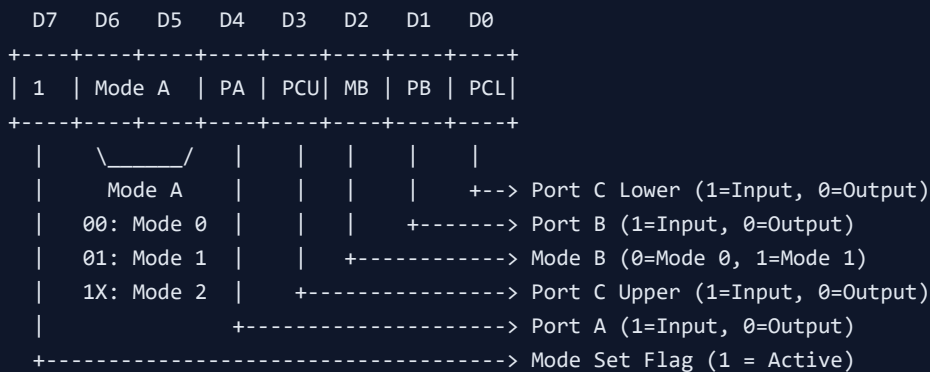


### 2. 8255 Port Selection Truth Table

| <u>\overline{CS}</u> | A <sub>1</sub> | A <sub>0</sub> | Selected Register / Port         |
|----------------------|----------------|----------------|----------------------------------|
| 0                    | 0              | 0              | Port A                           |
| 0                    | 0              | 1              | Port B                           |
| 0                    | 1              | 0              | Port C                           |
| 0                    | 1              | 1              | Control Word Register (CWR)      |
| 1                    | X              | X              | 8255 Deselected (High Impedance) |

### 3. Control Word Register (CWR) Formats

#### 1. I/O Mode Set Control Word (D<sub>7</sub> = 1)



#### 2. Operating Modes of 8255:

- Mode 0 (Basic I/O):** Simple input or output without handshake signals. All ports (A, B, C) operate independently.
- Mode 1 (Strobed I/O):** Handshake data transfer. Ports A and B transfer data using Port C pins for handshake control signals (STB, IBF, INTR).
- Mode 2 (Strobed Bidirectional Bus):** Port A acts as an 8-bit bidirectional bus using 5 pins of Port C for handshaking.
- BSR Mode (Bit Set/Reset Mode, D<sub>7</sub> = 0):** Used to set or reset individual bits of **Port C only** without affecting other bits.

### 4. Must-Write Points for Exams

- 8255 has 24 I/O pins configured into Port A, Port B, and Port C (Upper & Lower).
- D<sub>7</sub> = 1** in Control Word indicates **I/O Mode**; **D<sub>7</sub> = 0** indicates **BSR Mode**.
- Port C provides handshake status lines for Mode 1 and Mode 2 transfers.

### 5. Quick Recall Flow

8255 PPI -> 24 Pins (Ports A, B, C) -> Port Selection (A1, A0) -> Modes: Mode 0 (Basic), Mode 1 (Strobe), Mode 2 (Bi-dir), BSR Mode